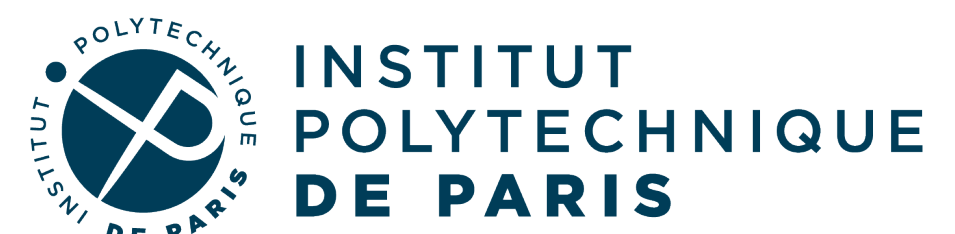


Understanding graph representation learning in RL-based logic synthesis

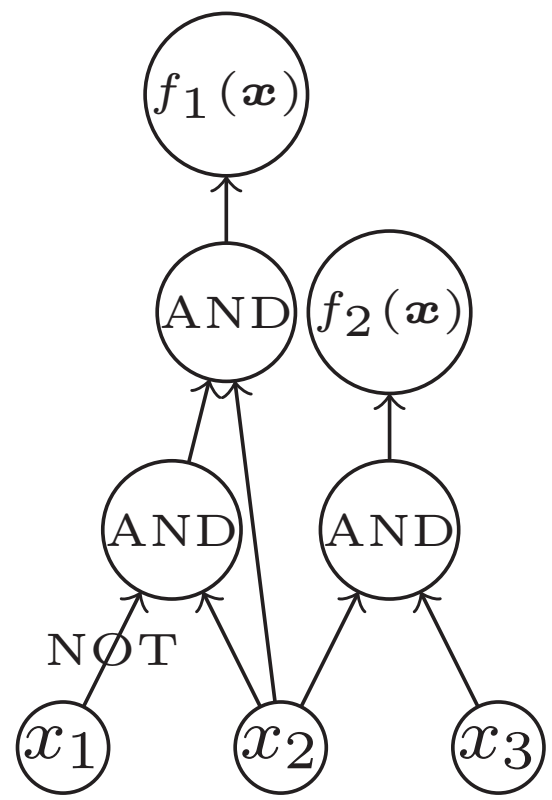
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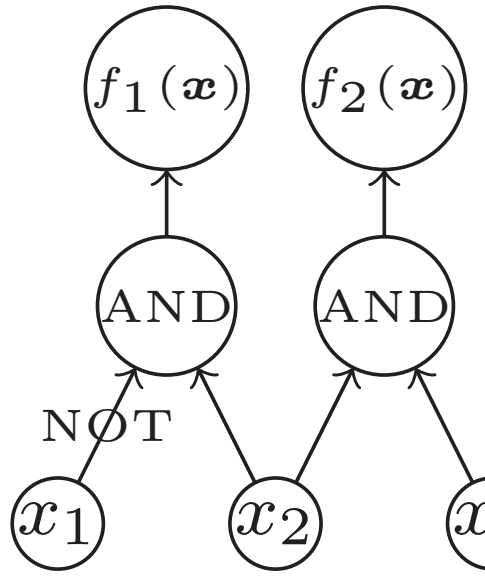


1. Logic synthesis with AIGs

Chip design needs the smallest circuit encoding a given Boolean function, which in practice means the smallest and-inverter graph (AIG): a DAG whose nodes are *AND* gates and whose edge labels are *NOT* signals. Below are two functionally equivalent AIGs of $f(x_1, x_2, x_3) = (\overline{x_1} \wedge x_2, x_2 \wedge x_3)$ with different numbers of nodes; *ABC* primitives rewrite an AIG locally to reduce the *AND* count while preserving equivalence (Mishchenko et al., 2006).



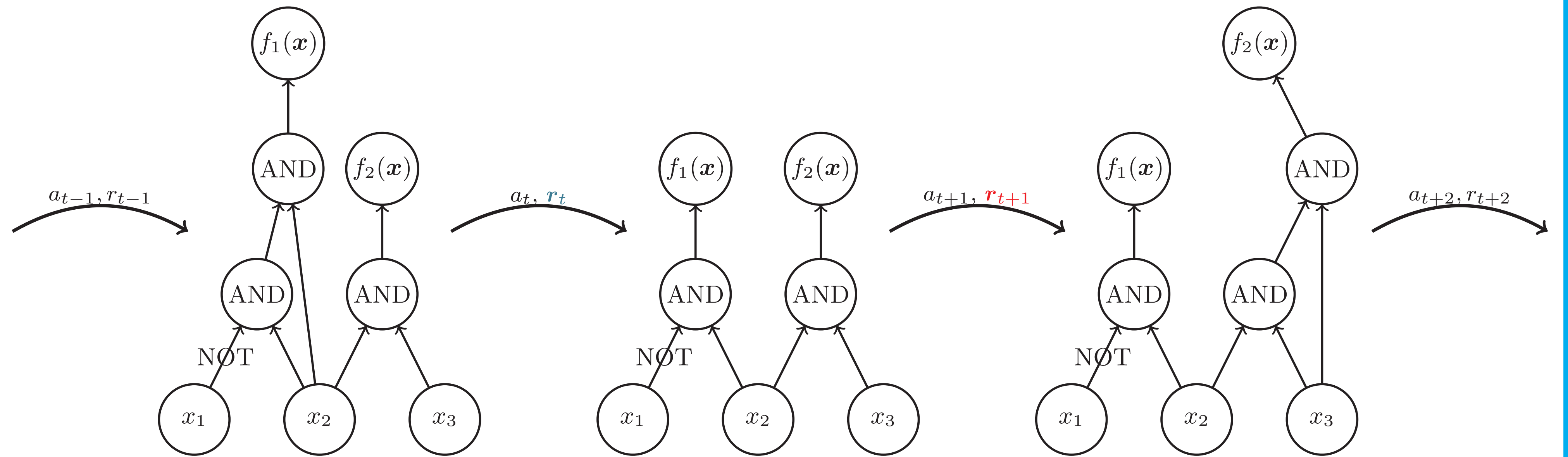
$\mathcal{G}_1$: 8 nodes, 4 levels.



$\mathcal{G}_2$: 7 nodes, 3 levels.

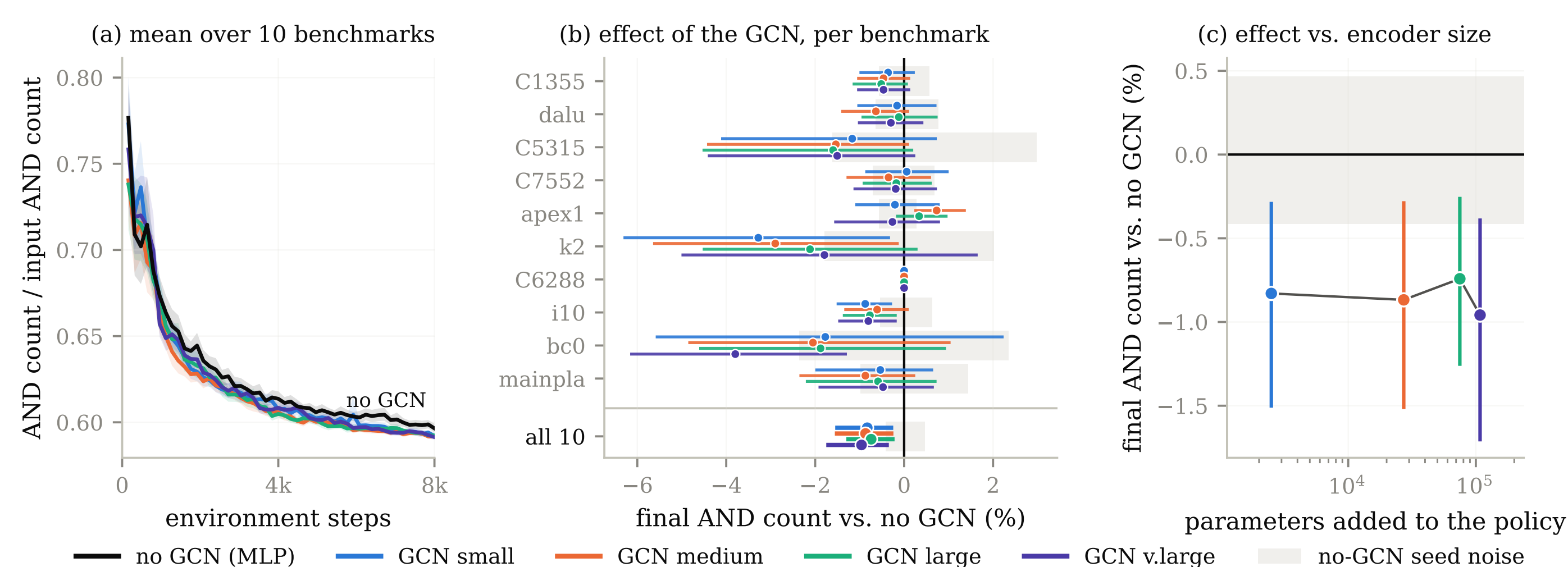
2. AIG reduction as an MDP

Each action is one *ABC* primitive applied to the whole graph and the reward grows as the number of *AND* nodes drops; since the primitives preserve functional equivalence, every state reachable from $\mathcal{G}_0$ still computes f . Most reinforcement learning (RL) and other machine learning-based searches for logic synthesis act at this granularity (Zhu et al., 2020; Hosny et al., 2020; Neto et al., 2023). DAG generative models fail to generate functionally equivalent AIGs Stoll et al. (2026).



3. Graph representation and RL

We train **tuned** PPO agents on the ten benchmarks of Zhu et al. (2020) and compare GCN policies against policies that only see graph statistics (current # nodes, # edges, last few graph edits taken, current time step). Adding a GCN changes the final *AND* count by about one percent whatever the size of the encoder, and per benchmark that change mostly stays inside the no-GCN policy's own spread over seeds (grey band).



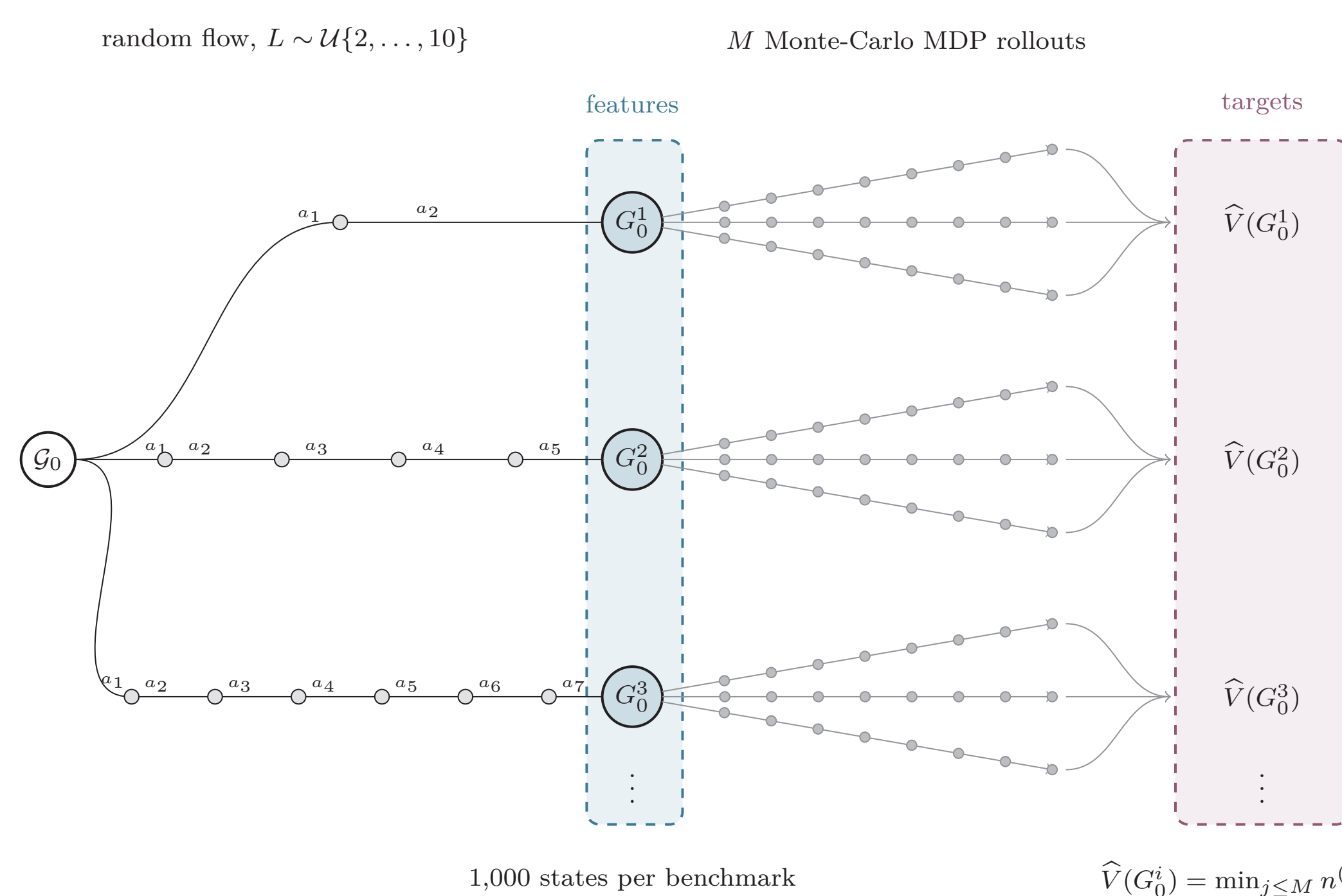
4. Final AIG sizes

Mean *AND* count after $H = 20$ *ABC* primitives, over the five seeds of each configuration's best hyper-parameter cell, $\pm$ one standard deviation. All five encoders land within a few nodes of each other, and all of them below the **resyn2** baseline.

AIG	Starting nodes	no GCN (MLP)	GCN small	GCN medium	GCN large	GCN v. large	resyn2
C1355	504	389 $\pm$ 3	387 $\pm$ 1	387 $\pm$ 0	387 $\pm$ 1	387 $\pm$ 0	390
dalu	1740	1008 $\pm$ 9	1006 $\pm$ 6	1001 $\pm$ 3	1006 $\pm$ 6	1005 $\pm$ 2	1052
C5315	1773	1303 $\pm$ 43	1287 $\pm$ 8	1283 $\pm$ 2	1282 $\pm$ 6	1283 $\pm$ 5	1309
C7552	2074	1371 $\pm$ 12	1372 $\pm$ 10	1367 $\pm$ 11	1369 $\pm$ 5	1369 $\pm$ 10	1455
apex1	2124	1066 $\pm$ 7	1064 $\pm$ 11	1074 $\pm$ 4	1070 $\pm$ 4	1063 $\pm$ 14	1179
k2	2190	1049 $\pm$ 26	1015 $\pm$ 32	1019 $\pm$ 27	1027 $\pm$ 21	1030 $\pm$ 36	1194
C6288	2337	1870 $\pm$ 0	1870 $\pm$ 0	1870 $\pm$ 0	1870 $\pm$ 0	1870 $\pm$ 0	1870
i10	2675	1715 $\pm$ 13	1700 $\pm$ 3	1705 $\pm$ 9	1702 $\pm$ 3	1702 $\pm$ 6	1829
bc0	3501	1512 $\pm$ 46	1485 $\pm$ 39	1481 $\pm$ 31	1484 $\pm$ 28	1455 $\pm$ 13	1713
mainpla	6359	2318 $\pm$ 37	2306 $\pm$ 14	2298 $\pm$ 11	2305 $\pm$ 23	2307 $\pm$ 14	2476

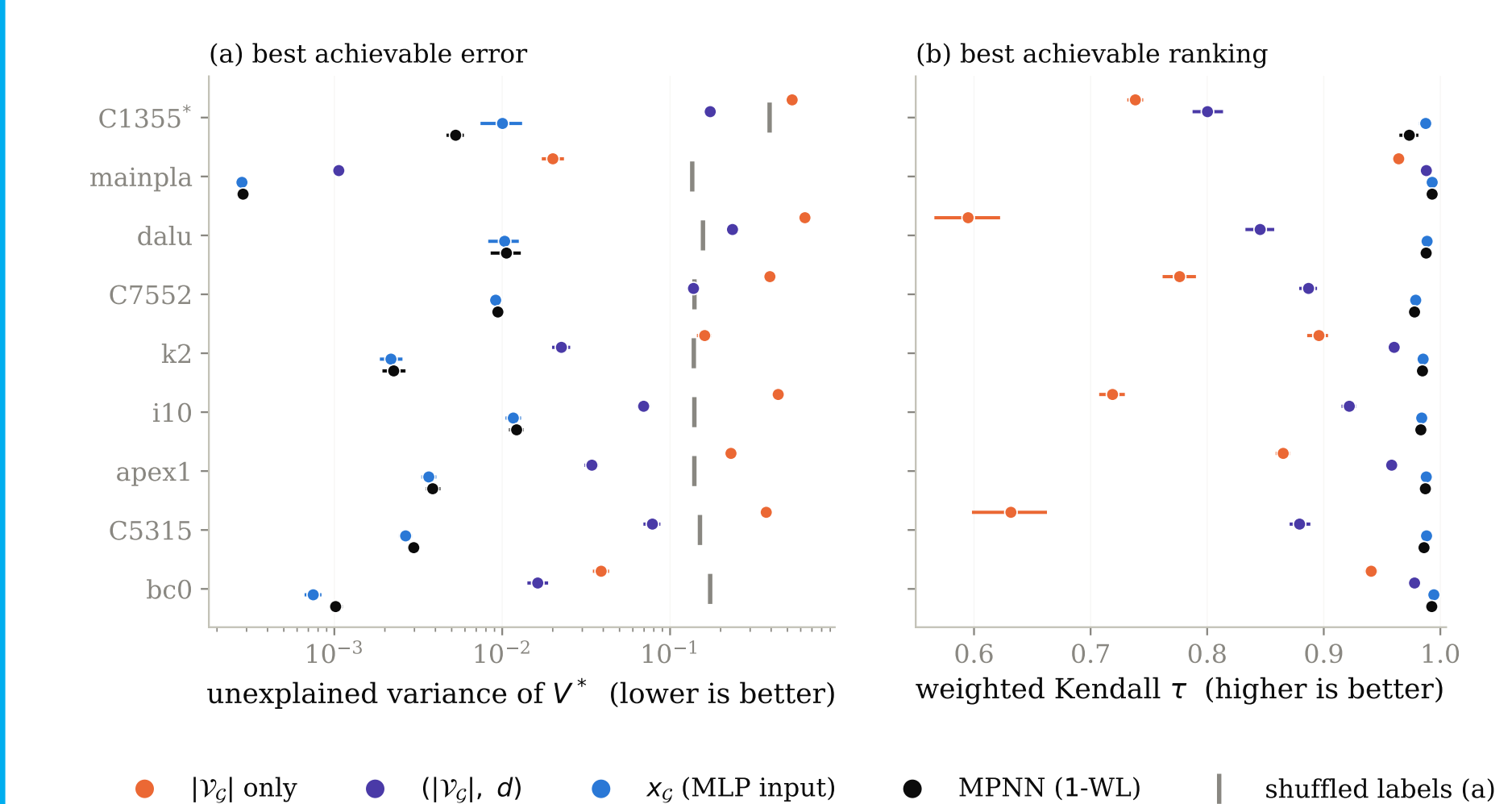
5. Why don't graph representations help?

Computing V^* exactly is out of reach, so we estimate it: a random flow of *ABC* primitives applied to a benchmark $\mathcal{G}_0$ gives a state G_0^i , whose target $\hat{V}(G_0^i)$ is the smallest *AND* count reached by $M = 100$ rollouts of eight further random primitives. We build this supervised dataset in order to study the predictive power of message passing networks (1-WL) on AIGs with **GraphTester** (Akbiyik et al., 2023).



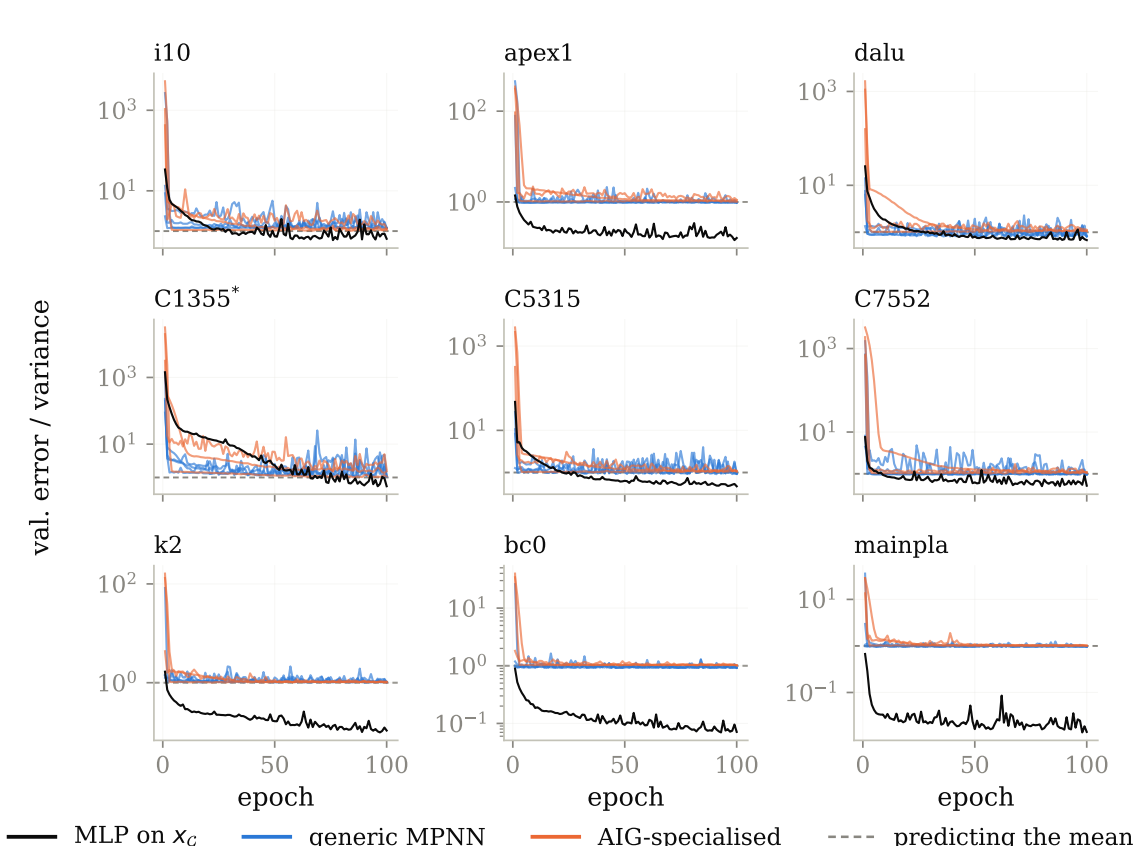
6. 1-WL representations for AIGs

Best error (a) and best ranking (b) achievable on $\hat{V}$ by each model class, over nine benchmarks and ten seeds: each class is replaced by the optimal predictor for what it can distinguish (Akbiyik et al., 2023), so no model in that class can do better. *C1355 has only six distinct targets.



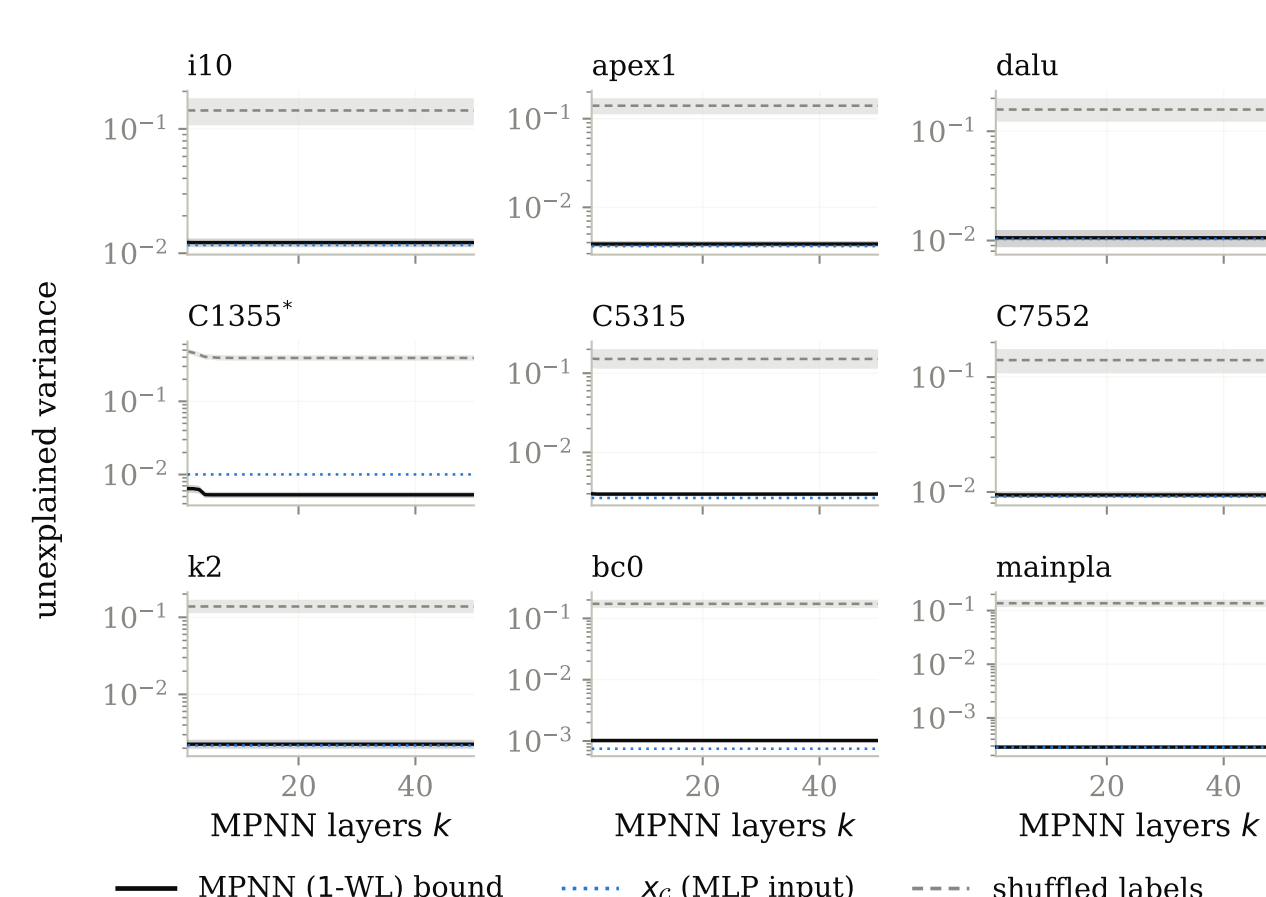
7. AIG-specific value models

Validation error over training when predicting $\hat{V}$ with tuned AIG-specific models (Li et al., 2022) and with generic graph representation models.



Additional results

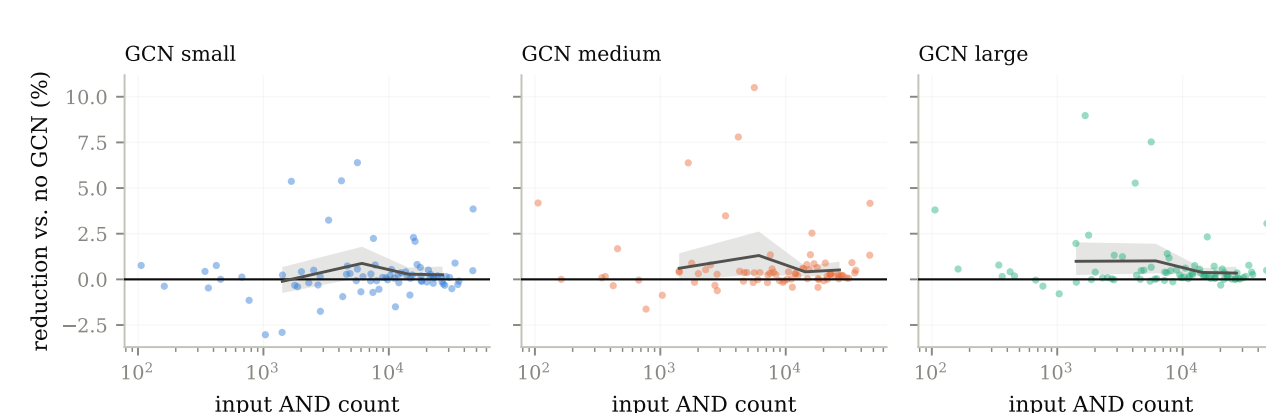
The message passing bound moves by at most 1.1×10^{-3} between $k = 1$ and $k = 50$ refinement iterations, and labeling edges with direction or with the complement bit changes it by less than 10^{-6} .



8. Future work

- Extend the analysis to non 1-WL representations, since some AIG-specific models might not be 1-WL.
- ABC* implements much more than 5 primitives, but no search method uses them all. More actions would mean more varied states, which could be where graph representation learning helps.

Reduction against the MLP as a function of input circuit size on **ex200-ex299**, the line being the mean over size quartiles.



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